

Saksham Sahu

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Portfolio: sakshamsahu.me | Faridabad, India

EDUCATION

Bennett University

Bachelor of Technology in Computer Science Engineering; CGPA: 8.72

Aug 2023 – Jun 2027

Noida, India

EXPERIENCE

DRDO, Ministry of Defence

Generative AI Intern

May 2026 - Jul 2026

- Engineered secure, locally hosted **Advanced RAG** system on a **Unix/Linux** environment using LLMs, vector databases and embedding models, achieving **92.5% retrieval accuracy** on a **460-question** benchmark over a confidential knowledge base
- Developed **Hybrid RAG** architectures with semantic search, keyword retrieval and reranking while reducing query latency by **~25%**
- Built scalable document processing pipelines handling **11,000+ pages**, including chunking, embedding generation, indexing, and structured **logging** for debugging, reducing hallucinations through multi-stage retrieval

Imarticus Learning

Data Analyst Intern

Jun 2025 - Jul 2025

- Led a team of **5 analysts** to develop **interactive dashboards** delivering key business insights to stakeholders
- Analyzed **50,000+ row** business datasets using **MySQL, Python and Power BI** to extract actionable insights
- Visualized insights using **Matplotlib and Seaborn** for weekly stakeholder reports

SKILLS

Languages : C++, Python, Java, JavaScript, TypeScript

Web & Backend : React.js, Next.js, Node.js, Express.js, REST API

Databases : MySQL, MongoDB, PostgreSQL, Vector Databases

Tools : GitHub, Postman, Power BI, Figma

Cloud & DevOps : AWS(ECR, ECS, EC2, ALB), Docker, Git, CI/CD pipelines

Core CS : Data Structures & Algorithms, OOPs, DBMS, Operating System, Computer Networks, System Design

Gen AI : RAG, LLMs, LangChain, LangGraph, Prompt Engineering, ChromaDB, Vector Search

PROJECTS

Parakh Trading Intelligence | NestJS, TypeScript, Kafka, TimescaleDB, Redis, Docker, OpenTelemetry

Sep 2026

- Developed an event-driven algorithmic trading platform using **9 microservices** and **Kafka**, decoupling real-time market data ingestion, backtesting and alert generation
- Built a **TimescaleDB** time-series pipeline for live indicator calculation (SMA, EMA, RSI, MACD), with **circuit breakers** for third-party API outages and **OpenTelemetry** tracing across HTTP services
- Reduced gateway **p95** latency from **~1.5s to 113ms** at 100 concurrent users (**~1,470 req/s, k6**) using **PgBouncer** pooling, HTTP keep-alive and short-TTL **Redis caching**; load balanced across 3 gateway replicas behind Nginx

Askra AI | React.js, FastAPI, FAISS, BM25, MongoDB, JWT, Docker, Grafana k6

Jun 2026

- Architected an enterprise **Agentic RAG** platform for private-document Q&A with a **7-layer pipeline** and LLM-as-judge self-correction, raising answer accuracy from **68% to 92%** over a baseline RAG on a ground-truth QA benchmark (LLM-judged)
- Engineered hybrid **FAISS + BM25** retrieval with cross-encoder reranking, running both searches concurrently via thread futures for **354ms** P95 TTFT; integrated a local OCR model to auto-process scanned PDFs
- Shipped async **FastAPI + MongoDB** backend with **JWT** auth, department-level **RBAC** and **SSE** streaming from Groq LLMs (**~100ms TTFT**); cut cold-start latency **83%** via startup model pre-warming; containerized with **Docker** and **Nginx**

Social-ish | React.js, Node.js, MongoDB, WebSocket, Gemini API

Apr 2025

- Built and deployed a **full stack social platform** for introverts with interest based matching and real time WebSocket chat
- Designed a scalable REST API with **JWT authentication** and MongoDB indexing for fast interest matching queries

ACHIEVEMENTS

Leetcode/amSaksham

Solved **700+ DSA problems**

Ongoing

Innovate 2.0: Hack to Build – IIIT Noida

National level Winner among 200+ teams

2024

TechArena 2025 Project Showcase

Secured 3rd position

2025